$$m_1 = \rho(S - S_0)h_0 \tag{12}$$

$$K = \frac{mv_0^2}{2} + \frac{m_1 v^2}{2} = \rho S_0 h_0 \frac{v_0^2}{2} + \rho(S - S_0)h_0 \frac{1}{2}\left(\frac{S_0}{S - S_0}v_0\right)^2 = \frac{1}{2}\frac{S_0 S}{S - S_0}\rho h_0 v_0^2 . \tag{13}$$

A2.3

$$\frac{1}{2}\frac{S_0 S}{S - S_0}\rho h_0 v_0^2 + \frac{1}{2}\frac{S_0 S}{S - S_0}\rho g x^2 = E = const . \tag{14}$$

$$T = 2\pi\sqrt{\frac{h_0}{g}} . \tag{15}$$

구분	평가 항목	배점	
A1	(1) $ma = mg - \rho S_0(h_0 + x)g$	0,2	**0,8**
	(2) $mg = \rho S_0 h_0 g$	0,2	
	(3) $a = -\frac{g}{h_0}x$	0,2	
	(4) $T = 2\pi\sqrt{\frac{h_0}{g}}$	0,2	
A2.1	(5) $\Delta U_1 = -mgx$	0,2	**1,2**
	(6) $S_0 x = (S - S_0)y \quad \Rightarrow \quad y = \frac{S_0}{S - S_0}x$	0,2	
	(7) $\Delta m = \rho S_0 x$	0,2	
	(8) $\Delta h_C = h_0 + \frac{1}{2}(x + y) = h_0 + \frac{1}{2}\left(x + \frac{S_0}{S - S_0}x\right) = h_0 + \frac{1}{2}\frac{S}{S - S_0}x$	0,2	
	(9) $\Delta U_2 = \Delta m g \Delta h_C = \rho S_0 x g\left(h_0 + \frac{1}{2}\frac{S}{S - S_0}x\right)$	0,2	
	(10) $\Delta U = \Delta U_1 + \Delta U_2 = \frac{1}{2}\frac{S_0 S}{S - S_0}\rho g x^2$	0,2	
A2.2	(11) $v_0 S_0 = v(S - S_0) \quad \Rightarrow \quad v = \frac{S_0}{S - S_0}v_0$	0,2	**0,6**
	(12) $m_1 = \rho(S - S_0)h_0$	0,2	
	(13) $K = \frac{mv_0^2}{2} + \frac{m_1 v^2}{2} = \rho S_0 h_0 \frac{v_0^2}{2} + \rho(S - S_0)h_0 \frac{1}{2}\left(\frac{S_0}{S - S_0}v_0\right)^2 = \frac{1}{2}\frac{S_0 S}{S - S_0}\rho h_0 v_0^2$	0,2	
A2.3	(14) $\frac{1}{2}\frac{S_0 S}{S - S_0}\rho h_0 v_0^2 + \frac{1}{2}\frac{S_0 S}{S - S_0}\rho g x^2 = E = const$	0,2	**0,4**